

ABDELRAHMANE BENSAHLA

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AI engineer shipping production LLM systems on the Claude API, TypeScript, Next.js, and PostgreSQL. Built and monetized an AI SaaS solo, from prompt pipeline to Stripe billing. Runs internal applications used daily by hundreds of employees.

EXPERIENCE

New Jersey Turnpike Authority

May 2025 - Present

Full Stack Engineer

Woodbridge, NJ

- Replaced legacy Microsoft Access workflows with three internal web apps used daily by hundreds of employees across 3 departments. Timed against the legacy system: 40% faster lookups.
- Cut topology page load from 2.4s to under 500ms by adding composite indexes on device-relationship tables and rewriting two N+1 join patterns in Drizzle.
- Eliminated 10+ hours of weekly manual work with Python and Node services for scheduled ETL, network monitoring, and Microsoft Graph directory automation.
- Designed the production PostgreSQL schema behind 20,000+ records and the dashboards on it: MUI DataGrid Pro with server-side pagination, Zod-typed forms, server-action uploads, PDF export.
- Shipped role-based access control (RBAC) and Playwright end-to-end coverage on every CRUD, filter, and export path. Deployed behind nginx; ships through PR review with QA and DevOps sign-off.

PRODUCTS

Job Cannon - AI resume-to-job matching SaaS | jobcannon.app

~30 paying subscribers

Next.js | TypeScript | Claude API | Clerk | Stripe | Neon | Drizzle | Resend | Vercel

- Built the LLM pipeline solo: Claude extracts a typed profile from an uploaded resume, a deterministic pre-filter narrows 5K nightly-scraped jobs, then Claude re-ranks the top 10 with reasoning.
- Enforce structured output with forced tool calls, Zod schema validation, and a single-retry fallback, so a malformed model response never reaches a subscriber's inbox.
- Score ranked results against source resumes each release and iterate on prompt structure from the failures, catching match-quality regressions before they ship.
- Held inference to ~2,000 tokens per user per day, roughly \$2 per subscriber per month against an \$8 price, sized deliberately for margin at scale.
- Chose keyword pre-filter plus LLM re-rank over vector embeddings: better precision at 5K jobs in a structured domain, with zero embedding infrastructure to maintain.
- Shipped the full subscription lifecycle: Stripe Checkout with trial, verified webhooks keeping Stripe in sync with Postgres, Customer Portal, Clerk auth, and a Vercel Cron digest job.

Sahl Solutions - Productized web design for local businesses | sahlagency.com

- Built a config-driven architecture where each client site generates from one typed `site.config.ts`, cutting per-client build from days to hours on a single upgrade path.

Job Fraud Classifier - Fraudulent job listing detection | github.com/abdelrahmanebensahla/jobfraudmodel

Python | TensorFlow / Keras | scikit-learn | pandas

- Trained a feed-forward neural network and a random forest to flag fraudulent postings across an 8,000-listing dataset, using TF-IDF text features and one-hot categoricals, scored on precision, recall, and F1.

SKILLS

AI / LLM: Claude API in production, prompt engineering, tool calling and forced structured output, typed extraction, output evaluation and regression scoring, cost and latency tuning, Claude Code, Cursor

ML: scikit-learn, TensorFlow / Keras, pandas, NumPy, TF-IDF and feature engineering, binary classification, precision / recall / F1 evaluation

Languages: TypeScript, JavaScript, Python, SQL

Web: React, Next.js (App Router, route handlers, server actions), Tailwind CSS, Material UI, MUI DataGrid Pro, Zod

Data & Infra: PostgreSQL, Drizzle ORM, Neon, query optimization and indexing, Docker, nginx, Vercel, GitHub Actions, Playwright, CI/CD

EDUCATION

Rutgers University, New Brunswick - B.S. Computer Science

Sep 2021 - May 2025